Project Title: Syria DryHeat Forecast: Machine Learning for One-Month-Ahead Heat and Drought Early Warning

1. Project Idea

- **Problem:** Syria faces recurring droughts and extreme heat that threaten agriculture and water supply, yet local decision-makers have no early warning. Our earlier work only described past trends (2000–2025) in six cities.
- **Goal:** Build a machine learning model that predicts whether the next month will be *high-risk (hot and dry)* in Damascus, Aleppo, Deir ez-Zor, Al-Hasakah, Latakia and Daraa and other Syrian cities.
- **Approach:** Use NASA POWER daily data (temperature, rainfall, humidity, wind, soil moisture, solar radiation, pressure) and engineered features (rainfall of previous months, number of very hot days, longest dry spell, anomaly versus the normal for the same month). Compare a historical-average baseline with Logistic Regression, Random Forest and XGBoost, then deploy the best model in a dashboard prediction page.
- **Users:** farmers, agricultural directorates and water managers, who can adjust irrigation and water storage a month in advance.

2. Relevance to SDGs

- **SDG 13 (Climate Action):** early warning and adaptation to climate extremes.
- **SDG 2 (Zero Hunger):** protecting crops and food security from drought.
- **SDG 6 (Clean Water):** supporting better water planning during dry periods.

3. Literature Examples

- **Li, Chang & Sherman (2025), "Forecasting Drought Using Machine Learning in California" (arXiv):** compares CNN, random forest, XGBoost and LSTM against a baseline persistence model to predict drought classification. It is similar to our plan of comparing several models against a simple baseline. https://arxiv.org/abs/2502.08622

- **Melese T, Assefa G, Terefe B, Belay T, Bayable G, Senamew A (2025**). **"Machine learning-based drought prediction using PDSI and TerraClimate data in Ethiopia" (PLOS ONE, 2025):** Random Forest performed best among the tested models, and soil moisture, precipitation and vapor pressure deficit were the most influential features. This supports our choice of tree-based models and of adding soil moisture to the features. https://doi.org/10.1371/journal.pone.0326174